

Amarnath D.J.

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SUMMARY

B.Tech Computer Science (AI & ML) student at SCTCE with hands-on experience building end-to-end machine learning pipelines, deploying production-grade web applications, and applying deep learning and computer vision techniques to real-world problems. Proficient in Python, Scikit-learn, TensorFlow, XGBoost, and Flask. Actively building expertise in data analysis, predictive modeling, and MLOps. Seeking roles in AI/ML Engineering, Data Science, Data Analytics, or Full-Stack Development.

SKILLS

AI & ML: Scikit-learn, TensorFlow, XGBoost, Regression, Classification, Clustering, Model Evaluation

Deep Learning & Computer Vision: YOLOv8, ByteTrack, CNN, OpenCV, Object Detection, Object Tracking

Data Science & Analysis: Pandas, NumPy, EDA, Data Wrangling, Statistical Analysis, Data Cleaning

Data Visualization: Matplotlib, Seaborn, Chart.js, Plotly

Languages: Python, Java, C, SQL, HTML

MLOps & Deployment: Flask, FastAPI, Streamlit, Docker, Render, AWS (basic), Azure (basic)

Tools: GitHub, Jupyter, VSCode, Notion

Data Structures & Algorithms: Arrays, Linked Lists, Stacks, Queues, Search and Sort Algorithms

Other: Object Oriented Programming, Operating Systems, MS Word, MS Excel

EXPERIENCE

Summer Intern

June 2026 – Present

ICFOSS, Karyavattom, Thiruvananthapuram

- Working on the **ML-Based Acoustic Rain Gauge** project, developing an acoustic rainfall estimation system using audio signal processing and machine learning.
- Predicting rainfall intensity directly from environmental audio recordings collected in real-world conditions.
- Synchronising model output with mechanical rain gauge measurements for ground-truth validation.

PROJECTS AND RESEARCH

Vision-Based Crowd Footfall Prediction System

2025–26

Tech: YOLOv8, ByteTrack, SARIMA, OpenCV, Python

- Built a real-time crowd monitoring system using YOLOv8 and ByteTrack for people detection and tracking.
- Applied SARIMA time-series forecasting to predict crowd flow from CCTV-derived data.
- github.com/Amarnathdj10/Vision-Based-Crowd-Footfall-Prediction

Pre-Match Player Rating Prediction using Machine Learning

2025–26

Tech: Python, XGBoost, Flask, Feature Engineering

- Built a full-stack ML application to predict football player match ratings using XGBoost and Flask.
- Engineered performance metrics including rolling form, opponent strength, and rest days.
- Deployed the model as a production-ready web application on cloud infrastructure.
- github.com/Amarnathdj10/Pre-Match-Player-Rating-Prediction

FIFA World Cup 2026 Predictor

In Progress

Tech: Python, Monte Carlo Simulation, Elo Ratings, Football Analytics, ML

- Building a tournament simulation and prediction engine for the 2026 FIFA World Cup.
- Incorporating team strength metrics, Elo ratings, squad valuation data, and Monte Carlo simulations.
- Estimating qualification probabilities and tournament winners through statistical modelling.
- github.com/Amarnathdj10/Fifa-World-Cup-2026-Predictor

Football Player Image Classification using Machine Learning

2024

Tech: Wavelet Transform, PCA, Logistic Regression, Flask, Render

- Developed and deployed an AI-based football player recognition web application on Render.
- Applied a feature-engineered ML pipeline combining Wavelet transforms, PCA, and Logistic Regression.
- github.com/Amarnathdj10

Machine Learning Foundation Projects (10+)

2023–2024

Tech: Python, Scikit-learn, SVM, Random Forest, Linear & Logistic Regression

- Implemented core ML algorithms from scratch — Linear Regression, Logistic Regression, SVM, Random Forest.
- Applied models to real-world datasets including housing price prediction and image classification.

EDUCATION

B.Tech in Computer Science – Artificial Intelligence & Machine Learning

Kerala

Sree Chitra Thirunal College of Engineering (SCTCE), Thiruvananthapuram

2023 – Present

ICSE / ISC Board (Class X & XII)

Kerala

Loyola School, Thiruvananthapuram

2008 – 2022

ACTIVITIES AND ACHIEVEMENTS

- KTU A-Zone Chess Championship 2025 – Runner-Up
- SCTCE Inter-Department Chess Championship – Champion, 2025
- SCTCE Chess Club Male Lead, 2025/26